

Central Counterparties and Risk

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Counterparty Credit Risk

What risk is clearing the 'solution' to?

Counterparty credit risk arises

- In contractual relationships, including derivatives, repo, and securities transactions...
- ...where a promise is made by one party at a point in time...
- ...that has to be fulfilled later...
- ...which they might fail to meet...
- ...and which, if they do not perform, could cause a loss to the other party

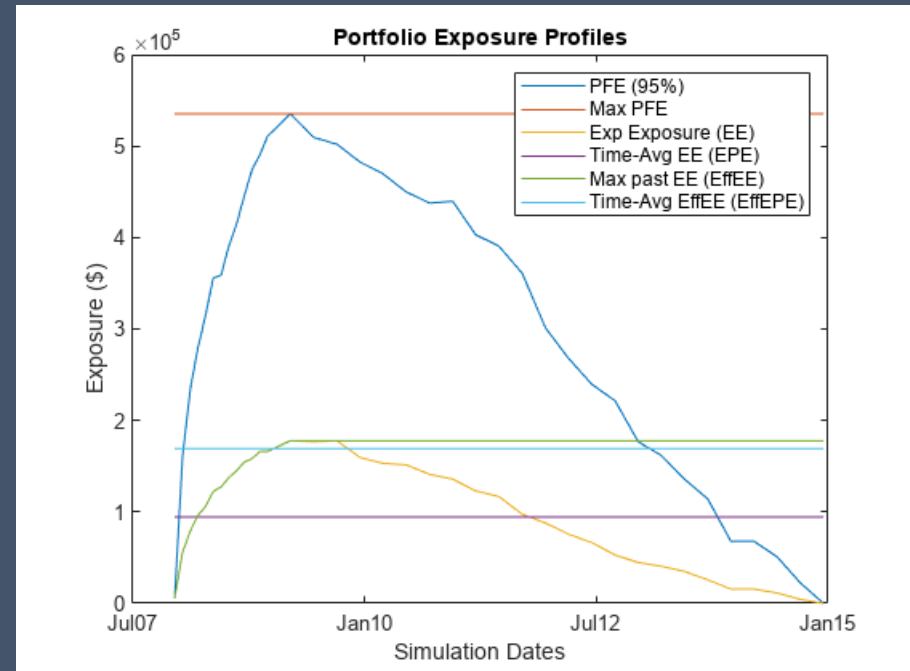
Paradigmatic Example

Interest rate swaps

- Often initially traded at or close to par
- But as time passes and rates move, the swap can become more valuable to one party
- And correspondingly less valuable to the other
- If the party who has lost money, on a mark to market basis, fails to perform
- Their counterparty loses a valuable asset

Counterparty credit risk profile

- Typically there is a time profile of exposure
- The more time passes, the more the market can move...
- ...but in a swap, the more time passes, the more coupons are paid
- So the **expected exposure**
$$E(\max(0, \text{replacement_value}))$$
- typically has a concave shape
- We can also study exposure at e.g. 95% confidence
- Usually by simulation of paths of market risk factors





Basic concepts in central clearing

Legal infrastructure

Modern financial markets rely on a **supportive legal system**

Key protections include:

- Enforceability of private law master agreements and rule books
- Novation of contracts
- (Close-out) netting of exposures
- Offset of collateral against exposures
- Perfection of collateral
- Safe harbour in bankruptcy for close-out
- Regulatory status of financial market infrastructures (FMI)
- Especially, settlement finality

Central counterparties

- A key type of modern FMI
- Often abbreviated to CCP
- The CCP sits between market participants, insulating each from the risk of the other



日本証券クリアリング機構
JAPAN SECURITIES CLEARING CORPORATION

CCP Rules

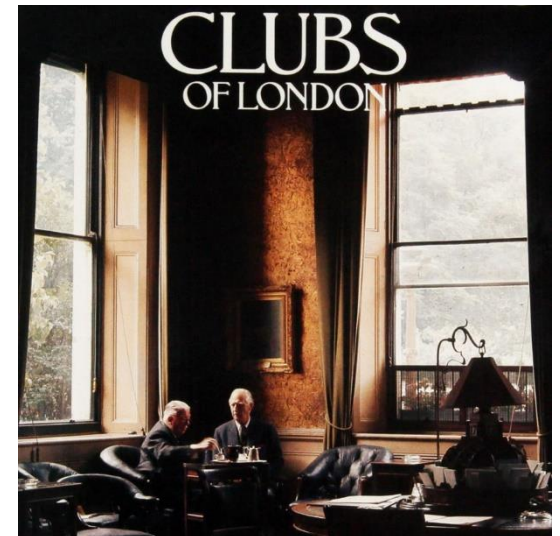
The direct participants of the CCP are called its **members**

As a condition of membership, they agree to adhere to rules:

- Their status, e.g. credit quality, net worth, regulation
- Their conduct in the market
- The specification of cleared contracts
- Posting margin and other resources

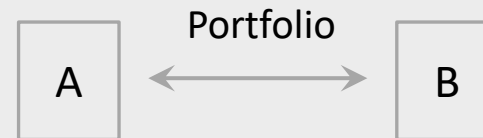
Further rules discussed later:

- Helping the CCP with default management
- Loss allocation
- Contract termination/tear-up

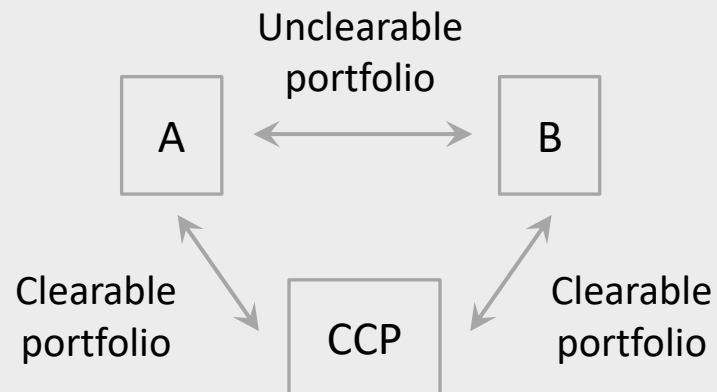


Central clearing

- In bilateral trading, there is a single contractual relationship

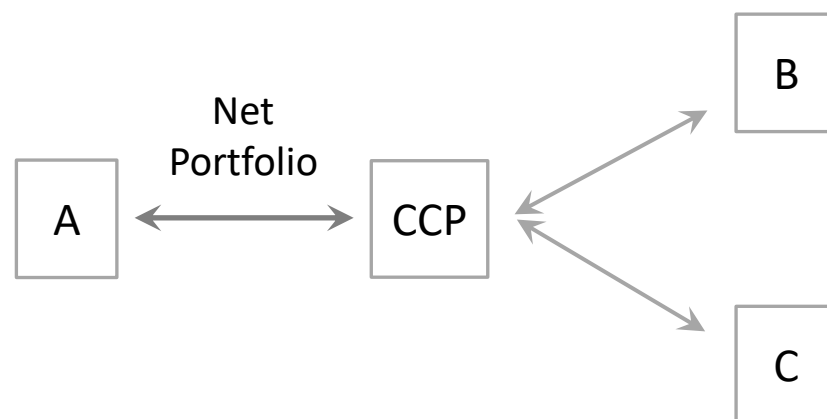
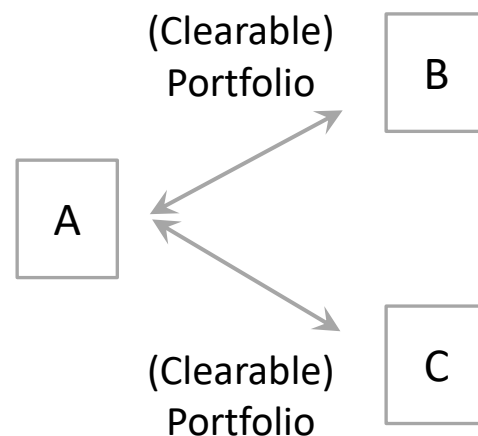


- Netting works for our purposes, so the basic unit of analysis is the **portfolio** between A and B
- This exposes A and B to each other's performance: they both have **counterparty credit risk**
- In central clearing, the contract is **novated** to a central counterparty or **CCP**



Netting

- The CCP does what it says in the name: it's **central**, and it is a **counterparty**
- Multilateral netting at the CCP reduces the credit exposure in the system compared to the counterfactual of bilateral trading



Two key issues in close-out

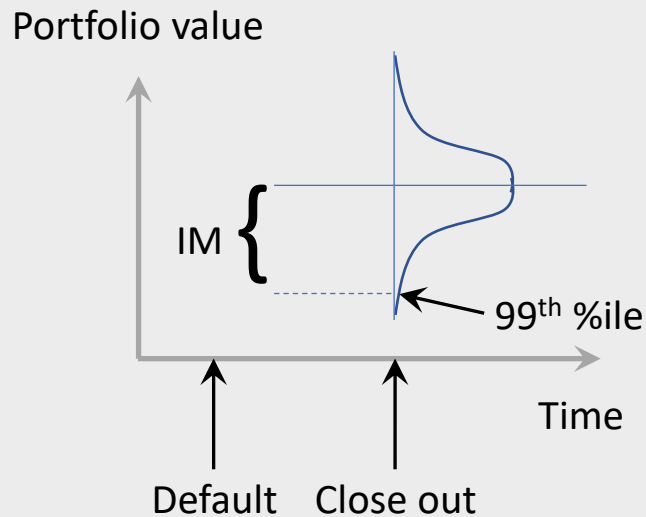
Stays in bankruptcy

- The automatic stay is triggered immediately on the filing of a bankruptcy petition in many legal systems
- It stops substantially all acts and proceedings against the debtor and its property
- Many financial contracts are unusual in that they are **exempt** from the stay
- This 'safe harbour' often applies to e.g. securities settlements, repo agreements and derivatives

Close-out netting

- The safe harbour means counterparties, including CCPs, can default manage **at once** after a default
- (Usually after a short notice period)
- They may also be able to declare a default before bankruptcy based on contractual arrangements, e.g. CCP rules
- Close-out netting means that the risk to be managed in default is the **net risk of the portfolio**

Initial and Variation Margin



- **Current exposure (or Mark to market)** is collateralised by **variation margin**
- When a default happens, closing out the portfolio does not happen immediately
- The **margin period of risk** ('MPOR') creates extra potential future exposure
- This is mitigated by **initial margin** ('IM')
- IM is typically calculated as some percentile of the distribution of exposure at the MPOR

Clearing members and default management

CCP rules typically require:

- Clearing members to participate in default management processes, e.g.
 - Bid in default auctions
 - Lend cash for default management
 - Test their capability in regular firedrills
- And pay for losses above margin created in default management
 - The obligation to contribute to the CCP's financial resources, e.g. the default fund
 - Various contingent obligations especially in recovery, i.e. if the CCP becomes stressed

Default management and auctions

- The key to understanding many aspects of clearing is **default management**
- The CCP must be able to
 - **Close out** the defaulter's portfolio
 - Liquidate its **collateral**
 - Absorb any losses using the waterfall
 - And re-resource itself if necessary
 - All while continuing to clear and to perform on its other obligations
- In listed products, it may be possible to close defaulter's positions in the market
- But for OTCs, a key tool is the **auction**
 - The CCP convenes a **default management group** of seconded traders
 - They either hedge the defaulter's portfolio, or value it without hedges
 - Details of the portfolio are sent to bidders
 - There may be a requirement on all clearing members in the service to bid, and/or strong incentives to do so
 - Bids are collected and analysed
 - Either the portfolio is allocated to successful bidders, or the auction fails

Continuity of provision of critical economic functions

Losses from default management and the default waterfall

- Typically CCPs call margin at least daily, and all the large ones can call for intra-day margin
- 99% (or 99.5%) of PFE isn't that secure, so there is a **default waterfall** of resources which can absorb tail losses caused by a defaulting clearing member
- **Skin in the game** ('SITG') is the CCP's own equity at risk
- The **default fund** ('DF') is a mutualised resource contributed by the CCP's members

Resource	Party Paying
(Recovery measures)	Clearing members
Default fund	Clearing members
Skin in the game	CCP
Defaulter's DF	Defaulter
Defaulter's margin	Defaulter



Where the money comes from to cover the cost of managing a default



How the default waterfall allocates losses

- The first mitigation of counterparty credit risk is margin, then (a wafer thin amount of) CCP capital, SITG, then mutualised resources
- Note that the prior bilateral model often had a large role for the survivor's **capital**
- Note too that margin has to be posted in a short period (e.g. two hours) after a call
- Whereas capital is a static resource



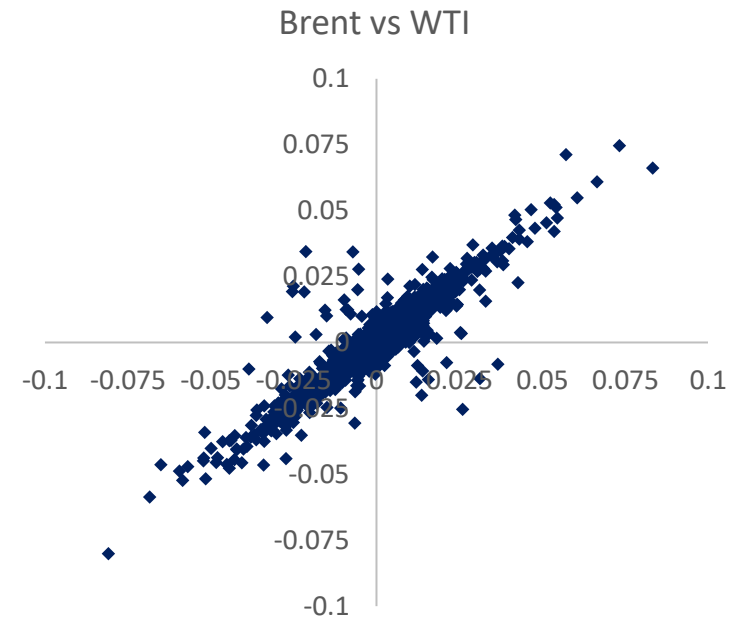
Margin requirements

Margin

- At least daily variation margin
- Intra-day VM when 'pre-defined thresholds' are exceeded
- Initial margin shall "be sufficient to cover losses that result from at least 99% of the exposures movements over an appropriate time horizon"
- 99.5% for OTC derivatives
- Minimum MPORs
 - 2 days for exchange traded derivatives in the house account
 - 1 day gross for the client account due to equivalence
 - 5 days for OTCs

Portfolio margining

- Risk factor returns are **co-move**
- This means that the total margin for a portfolio of instruments can be smaller than the sum of the margins for the individual instruments
- This is prudent if the co-movements exploited are robust in the tail
- EMIR and regulated regulation restricts the benefit of this portfolio margining to co-movements that are
 - Economically significant within an asset class and
 - Robust
- Portfolio margining is particularly effective for IRS as typical portfolios are large, have lots of closely offsetting tenors and may well have numerous currencies with significant positive co-movement

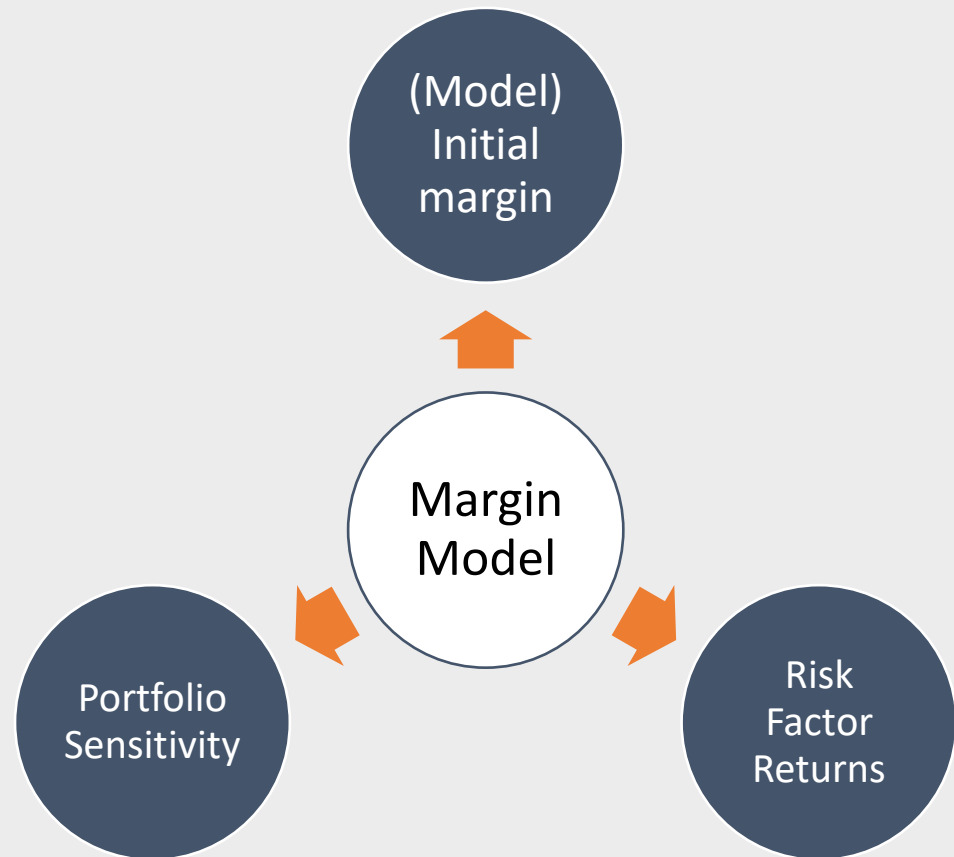


“A CCP may allow offsets or reductions ... across the financial instruments that it clears if the price risk of one financial instrument ... is **significantly and reliably correlated** ... with the price risk of other financial instruments”

Simple IM Model Architecture

Specification

- Modelled risk factors
- Confidence interval
- MPOR

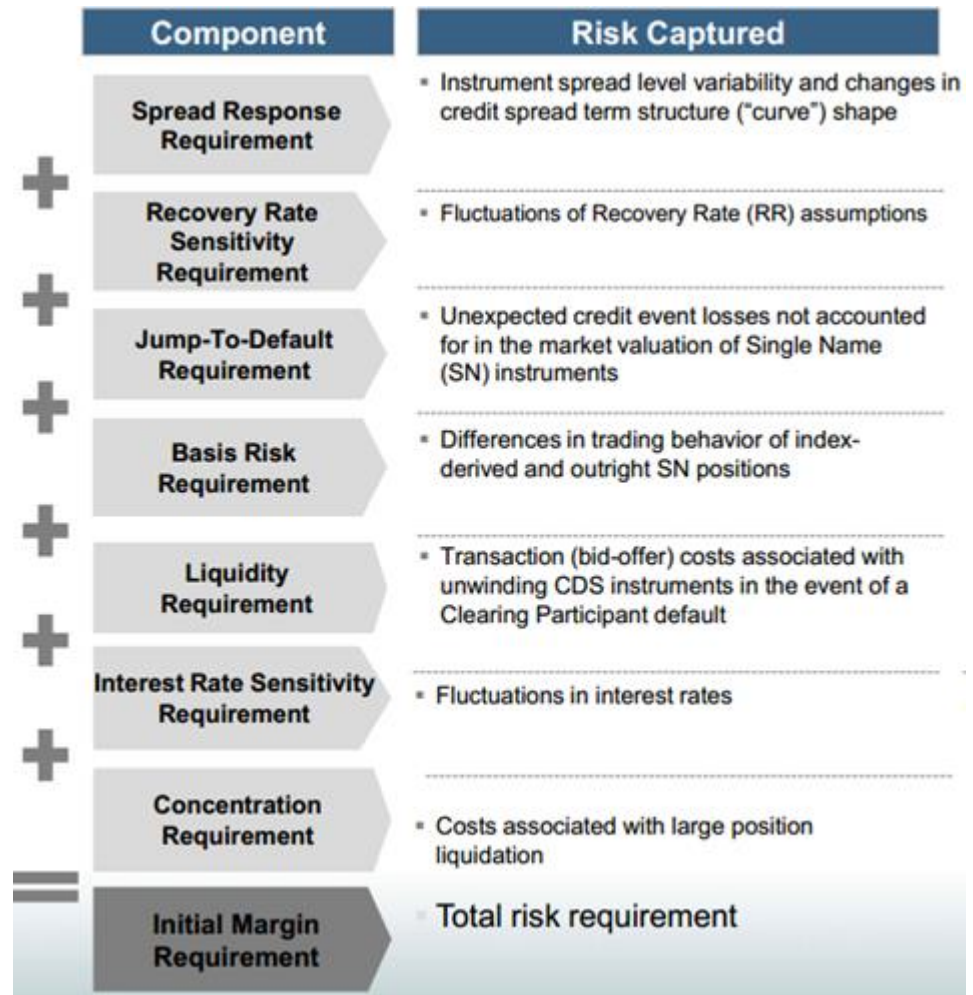


Simple example

- Product:
 - Interest rate swaps
- Modelled risk factors:
 - Swap rates along each curve
- Confidence interval:
 - 99.5% (regulatory minimum for OTC derivatives)
- MPOR:
 - 5 days
- Model Architecture:
 - Historical simulation value at risk
- Data window:
 - Five years (c.1250 observations)
 - So target percentile is c. 12th worst return
- Risk factor returns:
 - Log returns
- Portfolio representation:
 - Sensitivity to returns in each swap rate
 - Apply risk factor returns on each day, calculate P/L for that day

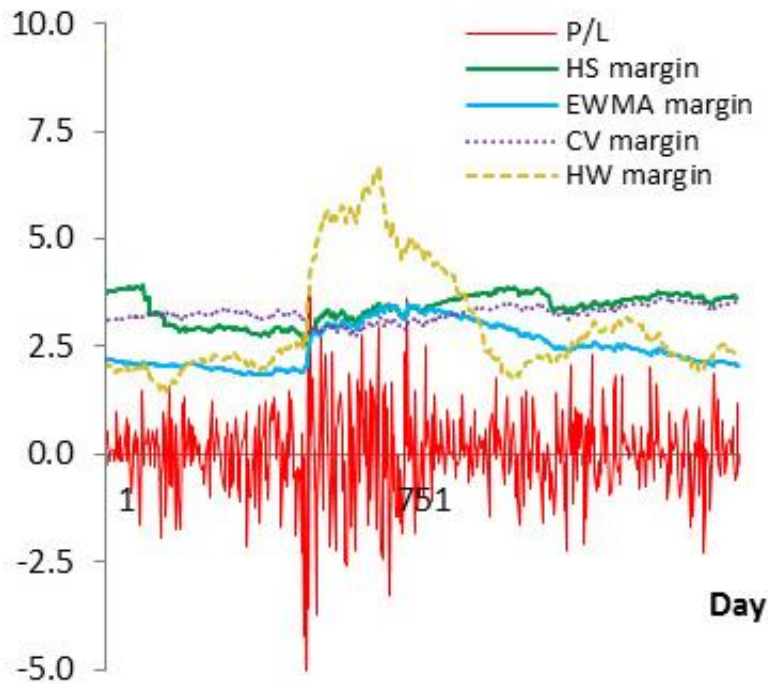
Non-model margin

- There are additional risks which are not (or usually not) captured in the CCP's core margin model, but which sometimes contribute to total margin requirements
- E.g.
 - Concentration risk
 - Member credit quality
 - Wrong way risk



Components of ICE CDS margin

Further Requirements



- A CCP shall ensure that the data used for calculating historical volatility **capture a full range of market conditions**, including periods of stress
- A CCP shall ensure that its policy for revising the confidence interval, the liquidation period and the lookback period deliver forward looking, stable and prudent margin requirements that limit **procyclicality**... This shall include avoiding when possible **disruptive or big step changes in margin requirements** and establishing transparent and predictable procedures for adjusting margin requirements...
- We return to this issue later

Verification of margin models

Two key requirements

- **Backtesting** – the statistical comparison of model outputs to actual P/L for a range of portfolios using historical market moves
 - A initial margin model targeting a 99% confidence interval should have an exemption roughly 5 days in 500
 - More sophisticated tests looking at clustering of exceptions, multiple confidence intervals, etc. are recommended
- **Sensitivity analysis** – whereby the sensitivity of margin to margin model parameters is tested to ensure they remain appropriate
 - c



Margin
payments

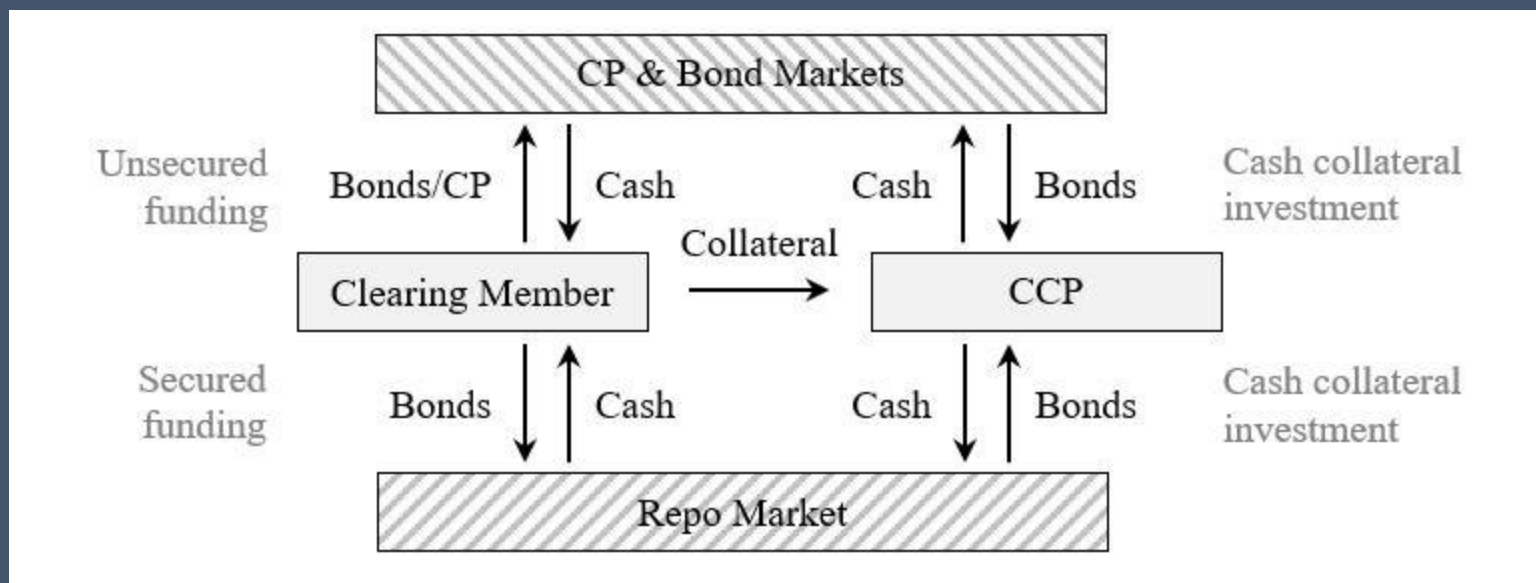
Margin Payments

- Variation margin is usually paid in **cash**
- Initial margin can often be paid in **securities**
- Requirements vary from CCP to CCP, but high quality government securities are usually acceptable
- Securities are often subject to a **haircut**, reflecting their potential decline in value between default and liquidation



The collateral system

- The system by which cash and non-cash collateral flows, margin is posted, and cash is transformed into securities and vice versa involves CCPs, bilateral counterparties, securities markets and prominently **the repo market**



Cash, securities and concentration

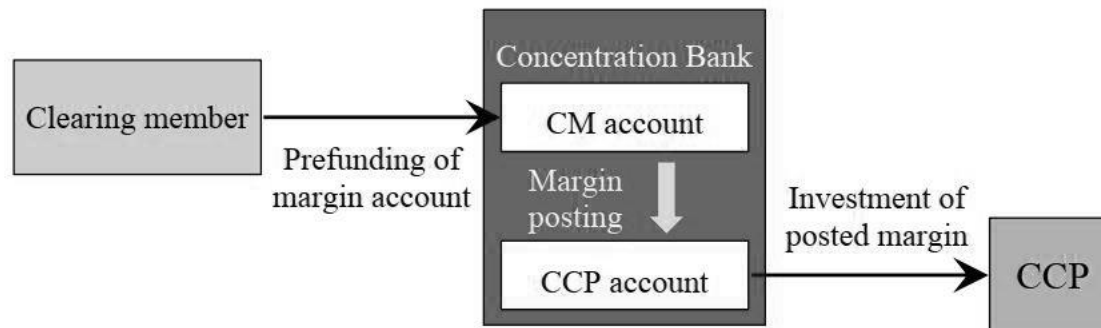
- There is a tension between cash and security collateral
 - Cash is the ultimately liquid asset
 - But it has to be invested, and this places additional requirements on the CCP
 - In particular, this investment needs to be easy to liquidate quickly
 - Also, some clients in particular may not have cash readily to hand
 - And securities are (much) easier to tag as client property, and thus to reduce clients' exposure to the default of their clearing member
- CCPs need to manage the risk that they have a concentrated position to manage in default or otherwise
- This is a key vulnerability
- Concentration can affect, e.g.
 - The cleared position itself
 - Securities in the collateral portfolio
 - Securities CCPs invest cash in
 - Their exposure to other counterparties, such as settlement banks

Right- and wrong way risk

- The usual idea is that collateral should be an **informationally insensitive asset**
- Are there any?
- **Right way risk** occurs when the collateral value goes up if the exposure goes up
- **Wrong way risk** occurs when the collateral value goes down if the exposure goes up
- Or when the probability of default of the counterparty goes up

Collateral posting

- The question of how collateral is posted is important in determining the risks in default
 - How much collateral will you get back if you are over-collateralised?
 - When will you get it back?
 - In client clearing, do you need to double fund a ported position for some period?
 - What is your exposure to the performance of intermediaries such as custodians?
- The two models are title transfer and pledge
- CCPs typically prefer title transfer models, as they then have unfettered access to the collateral



Title transfer

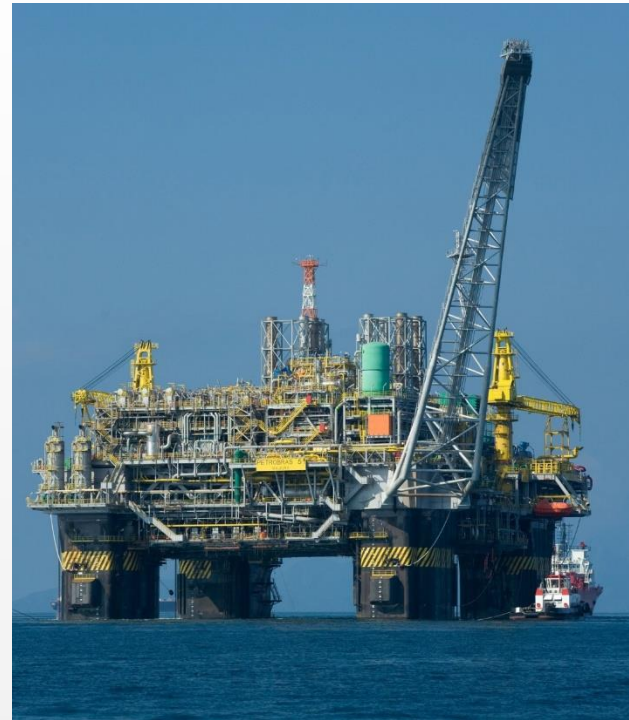
- When securities delivered, they become the recipient's property
- The recipient does *not* hold them in custody for the transferor
- The recipient typically only has an obligation to redeliver an equivalent security
- Client money protections do not apply

Pledge

- The transferor delivers the securities to the recipient
- They are held in trust for the transferor subject to a **security interest** securing their obligation
- Under the US version, the pledgee may sometimes be entitled to sell the pledged asset absolutely, under a process known as **rehypothecation**

Margin as liquidity risk

- Suppose you own an oil field, and have hedged the production with a short futures position
- You are (assuming you get the quantity right and match your production schedule and don't have basis risk and...) price hedged
- But suppose the oil price rises
 - Futures price falls
 - Variation margin call so you have to pay the CCP
 - But you haven't sold the oil for more yet, because it is still in the ground
 - Your hedging strategy has liquidity risk



Margin as liquidity risk - procyclicality

- Initial margin models are usually risk-based, so margin requirements rise in stress
- This is called 'procyclicality', and EMIR has (odd) requirements for anti-procyclicality (APC) tools. But...

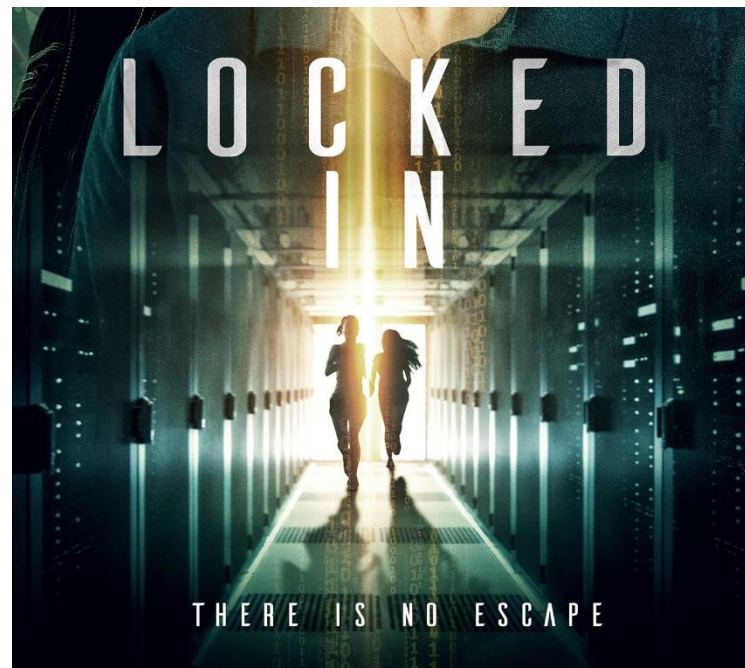
Given the rapid increases in market volatility experienced in March 2020, there was a broad-based and rapid increase in margin calls across the financial system. The aggregate changes in stocks and flows of margin differed in size across markets and CCPs.

Daily CCP VM calls were large, and significantly higher than the average flows observed between January and February 2020, increasing from around \$25 billion to a peak of \$140 billion, as based upon our survey results. These flows were a direct result of significant shifts in market volatility during the period (VM flows are directly determined by the realised mark-to-market changes in portfolios). Furthermore, while centrally cleared VM calls were predominantly made on an end-of-day (EoD) basis, there were significant intraday VM calls during the most stressful period in March. Most of these were made on pre-defined schedules, with some notable ad hoc calls on peak days.

The total IM requirement across CCPs increased by roughly \$300 billion over March 2020, with a further increase in excess collateral of \$115 billion, resulting in an overall increase in collateral pre-positioned at CCPs of \$415 billion (a roughly 40% increase relative to the average in February 2020).

Policy

- Work on CCP margin models: BCBS CPMI IOSCO, *Transparency and responsiveness of initial margin in centrally cleared markets*
- Work on users' liquidity management: FSB *Liquidity Preparedness for Margin and Collateral Calls: Consultation report*



- Margin simulation tools
- Disclosure:
 - Model parameters
 - Model design, inc. APC tools
 - Model responsiveness
 - Add-on procedures
 - Actual margin levels
- Governance framework

Multiple tensions in procyclicality policy

- CCPs v. clearing members v. clients
- CCP competition v. liquidity risk
- Cost v. liquidity risk
- Changes in market dynamics v. noise
- The role of the OTC market



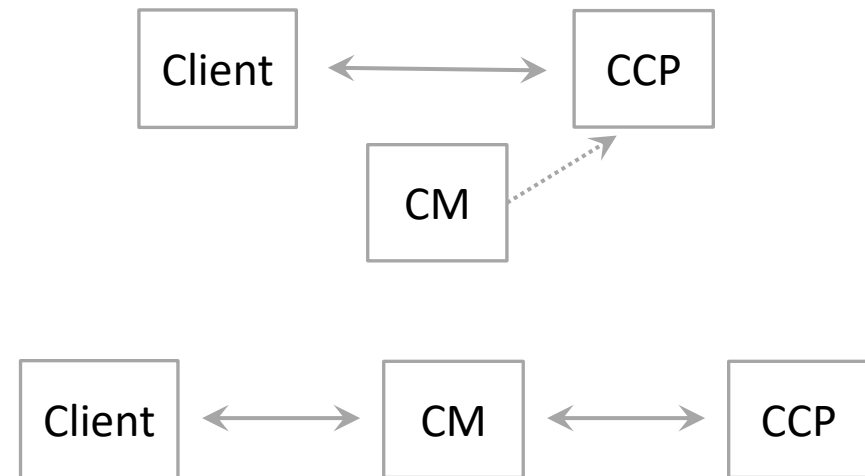


Segregation and portability

Client clearing

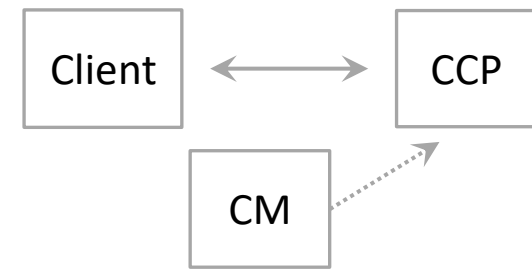
- Not all firms who want to clear can (or want to) meet CCP membership requirements
- Client clearing allows them to participate in the cleared market
- Here a clearing member either guarantees their performance to the CCP, in what is known as **agency model client clearing**
- Or intermediates them, in **principal to principal model client clearing**

- Clients are not required to post default fund contributions
- Clients can be **ported** if their clearing member fails – their positions are transferred to a surviving clearing member



Client clearing

- Client can have exposure to their clearing member
 - Counterparty credit risk on positions net of margin
 - Excess margin, if not posted to the CCP in the name of the client
 - Sometimes settlement exposures
 - Contract continuity
- Client clearing volumes are now significant



The split of initial margin between house and client accounts at five large OTC derivatives CCPs

In per cent



Reported as at end December 2017. ICEU = ICE Clear Europe. ICC = ICE Clear Credit. JSCC = Japan Securities Clearing Corporation.

Source: CPMI and IOSCO quantitative disclosures.

Segregation

- The name assets are held in matters if there is a default
- **Segregation** refers to the labelling of assets as for the benefit of someone else
- EMIR requires at least
 - That clearing member house accounts are separated from those of clients
 - That clearing members offer and CCPs support, at least, the choice between omnibus client segregation and individual client segregation
- There is a tension between the benefits of segregation and the costs

Omnibus vs. individually segregated accounts

- In Europe futures markets developed with all clients in one **omnibus** account
- The clearing member's own trades were in a separate **house** account
 - Clients were not separated from each other
- This was operationally efficient but risky if the clearing member failed and one or more clients were in deficit: there was **fellow client risk**
- EMIR requires that a higher level of segregation is offered, but not that it is affordable
- In practice CCPs and clearing members often offer a variety of models between omnibus and full segregation, as we show in the following examples from ICE
- These offer clients a choice of levels of risk, operational convenience/burden and cost

Portability

- In exchange traded markets, clients could sometimes have their position and collateral **ported** to another clearing member if theirs failed
 - E.g. a 'block' port of Refco's clients in 2005 to the purchaser (Man Group)'s FCM when they acquired the clearing business in bankruptcy
- Higher levels of segregation aid but do not guarantee porting

The cautionary tale of MF Global

- Refco failed due to a corporate fraud that did not affect the segregation of customer funds. But Refco's clients were not out of danger yet
 - Refco's purchaser was Man Group, and Man Group's brokerage arm was at the time called MF Global ('MFG')
 - Man floated MFG in 2008 and sold the last of its stake in 2009
 - On October 31st, 2011 MFG filed for bankruptcy, in what was at the time the 8th largest bankruptcy in US history, after a liquidity crisis
 - In the immediate aftermath, it was announced that there was a \$1.6B deficiency in segregated customer funds*
 - It appeared that MFG's management had used customer funds to meet the firm's liquidity needs

*The story of how much of this \$1.6B (out of c.\$5.5B total) was really deficient is complex and relates in part to a now-eliminated CFTC rule on the treatment of 'foreign' futures margin.

The Trustee Reports

“The investigation to date has found that transactions regularly moved between accounts and that funds believed to be in excess of segregation requirements ... were used to fund other daily activities of MF Global. In the past, such transfers were in amounts of less than \$50million, but as liquidity demands increased ... much larger amounts were used, apparently with the assumption that funds would be restored by the end of the day. By Wednesday, October 26, as the result of increasing demands for funds or collateral throughout MF Global, funds did not return as anticipated... [A] lack of intraday accounting visibility existed, caused in part by the volume of transactions being executed, and the ... U.S. segregated commodity customer account appears to have reached a deficit condition on Wednesday, October 26 that continued through to MF Global’s bankruptcy.”

- Segregation needs to work all the time to be effective
- The trustee eventually recovered all of the funds...
- But it took until 2014
- Corzine, the CEO, settled with CFTC for a \$5M fine and a lifetime ban after the DOJ declined to prosecute



CCP organisation and governance

Different models

- DTCC is the operator of the U.S. cash market securities CCPs—National Securities Clearing Corporation (NSCC) and Fixed Income Clearing Corporation (FICC)
- DTCC is **owned and governed by the users of its subsidiaries**, “all of whom commit capital as owners and pay fees for services”
- Its Board “is responsible for providing direction to and overseeing ... in the interests of the Corporation, its shareholders and other stakeholders including investors, issuers and participants in the financial markets which DTCC serves
- CME was founded in 1898 as a not-for-profit corporation. In 2000, CME demutualized and became a **shareholder-owned corporation**
- “As a consequence, we adopted a for-profit approach to our business, ... aimed at optimizing contract volume, efficiency and liquidity”
- Businesses
 - Exchanges
 - Trading platforms
 - “The company also offers clearing and settlement services across asset classes for exchange-traded and over-the-counter derivatives through its clearing house”

Risk Committees

- EMIR envisages a role for users in the governance of CCPs:
 - A CCP shall establish a risk committee, which shall be composed of representatives of its clearing members, independent members of the board and representatives of its clients
 - The risk committee shall advise the board on any arrangements that may impact the risk management of the CCP, such as a significant change in its risk model, the default procedures, the criteria for accepting clearing members, the clearing of new classes of instruments, or the outsourcing of functions
 - A CCP shall promptly inform the competent authority of any decision in which the board decides not to follow the advice of the risk committee



CCP financial
resources
beyond margin

CCP Equity Requirements and Skin in the Game

- EMIR sets minimum capital requirements for CCPs
- It starts strongly

“A CCP’s capital, including retained earnings and reserves, shall be proportionate to the risk stemming from the activities of the CCP”
- But then...

“It shall at all times be sufficient to ensure an orderly winding-down or restructuring of the activities over an appropriate time span and an adequate protection of the CCP against credit, counterparty, market, operational, legal and business risks which are not already covered by specific financial resources”
- So it usually ends up being six months of expenses
- Minimum skin in the game (SITG) is 25% of the capital requirement
- The intent is that SITG requirements incentivise good risk management at CCPs as the CCP’s own capital is at stake before that of the members
- However the SITG minimum is typically a small amount
 - So some clearing members are pressing for larger amounts to be required
- There *are* larger amounts in the groups of most large CCPs, they
 - E.g. Nasdaq shareholder’s funds, \$5.9B from their 2017 annual report
 - But only (as we will see) € 7M as SITG in one of its clearing services

Default fund sizing 1

- The default fund shall at least enable the CCP to withstand, under extreme but plausible market conditions, the default of the two clearing member to which it has the largest exposures
 - Cover 2 standard
 - The CCP has a suite of stress tests to determine the extreme but plausible moves
 - These are both historical and
 - Hypothetical
- Reverse stress testing

Eurex's Historical scenarios

1987 Black Monday

1989 DAX Move

2001 Twin Towers Attack

2001 Capital Market Move

2003 U.S. Housing Bubble

2008 Lehman Default

2008 Global Financial Crisis

2010 Flash Crash

2011 Fukushima Meltdown

2011 U.S. Down Grade

2011 SNB Intervention

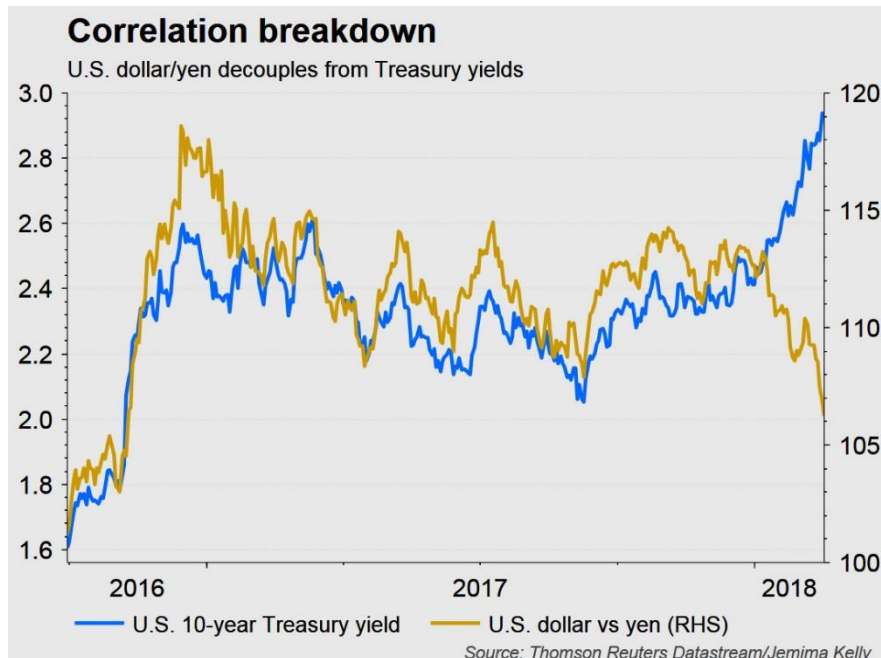
2012 ECB IR Event

2013 Cyprus Financial Crisis

2016 Brexit Referendum

Scenario Requirements

- Historical scenarios must including periods of extreme market movements observed over the past 30 years, or as long as reliable data have been available, that would have exposed the CCP to greatest financial risk
- The framework shall also consider, quantitatively and qualitatively, the extent to which extreme price movements could occur in multiple identified markets simultaneously
- The framework shall recognise that historical price correlations may breakdown in extreme but plausible market conditions



Default fund sizing 2

- For each account:
 - For each account, apply the scenario
 - Discard profits, take losses
 - Add the results for each clearing member
- This is their **stressed loss over IM**, or SLOIM
- The two largest are the 'cover 2' members
- And the default fund should be at least the sum of these two largest SLOIMs
- Each CM pays pro-rata, often based on their total IM

CPMI IOSCO disclosures

- Requirements for quantitative disclosures from CCPs on a quarterly basis
- The disclosures are intended to support the objectives of enabling stakeholders to compare CCP risk controls, including their financial condition and financial resources to withstand potential losses
- These are a valuable resource

4. Credit risk (Principle 4)

4.1	Total value of default resources (excluding initial and retained variation margin), split by clearing service if default funds are segregated by clearing service [see explanatory notes], split by: (a) pre-funded i. own capital that forms part of the default waterfall (further split by whether used before, alongside, or after, member contributions) ii. aggregate participant contributions (both amount required and post-haircut amount posted, where different) iii. other (b) committed i. own/parent funds that are committed to address a participant default (or round of participant defaults) ii. aggregate participant commitments to address an initial participant default (or initial round of participant defaults) iii. aggregate participant commitments to replenish the default fund to deal with a subsequent participant default (or round of participant defaults) after the initial participant default (or round of participant defaults) has been addressed
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Example – Swapclear

▪ Prefunded resources before (i.e. SITG)	£ 39,380,000
▪ Default fund	£ 5,000,000,000
▪ Cash at central banks	€ 1,319,320,000
▪ Secured cash including repo	€ 6,558,550,000
▪ Unsecured cash	€ 19,870,000
▪ Peak SLOIM in the last year for the largest member	£ 2,751,520,000
▪ House initial margin	£ 45,505,000,000
▪ Client initial margin (gross)	£ 49,407,360,000
▪ Client initial margin (net)	£ 4,179,340,000
▪ Margin model confidence interval	99.7%
▪ Type of model	Expected shortfall

Liquidity

- A CCP needs to be able to make payments when due, even during default management, e.g.
 - Changes in margin including withdrawals of excess margin or cash/non-cash substitutions
 - Payments on portfolio assets (e.g. swap coupons)
 - Securities settlements
- The risk that this will not be possible, or only at excessive cost, is **liquidity risk**
- EMIR requires CCP have a robust liquidity risk framework to manage this

Elements of liquidity plans

- Managing and monitoring, at least on a daily basis, liquidity needs across a range of market scenarios
- Maintaining sufficient liquid financial resources to cover liquidity needs and distinguish among the use of the different types of liquid resources
- The daily assessment and valuation of the liquid assets available
- Identifying sources of liquidity risk and the processes for liquidity shortfalls
- Assessing timescales over which liquid financial resources should be available
- Considering potential liquidity needs stemming from clearing members' ability to swap cash for non-cash collateral
- The replenishment of any liquidity the CCP may employ during a stress event

Sources

vs.

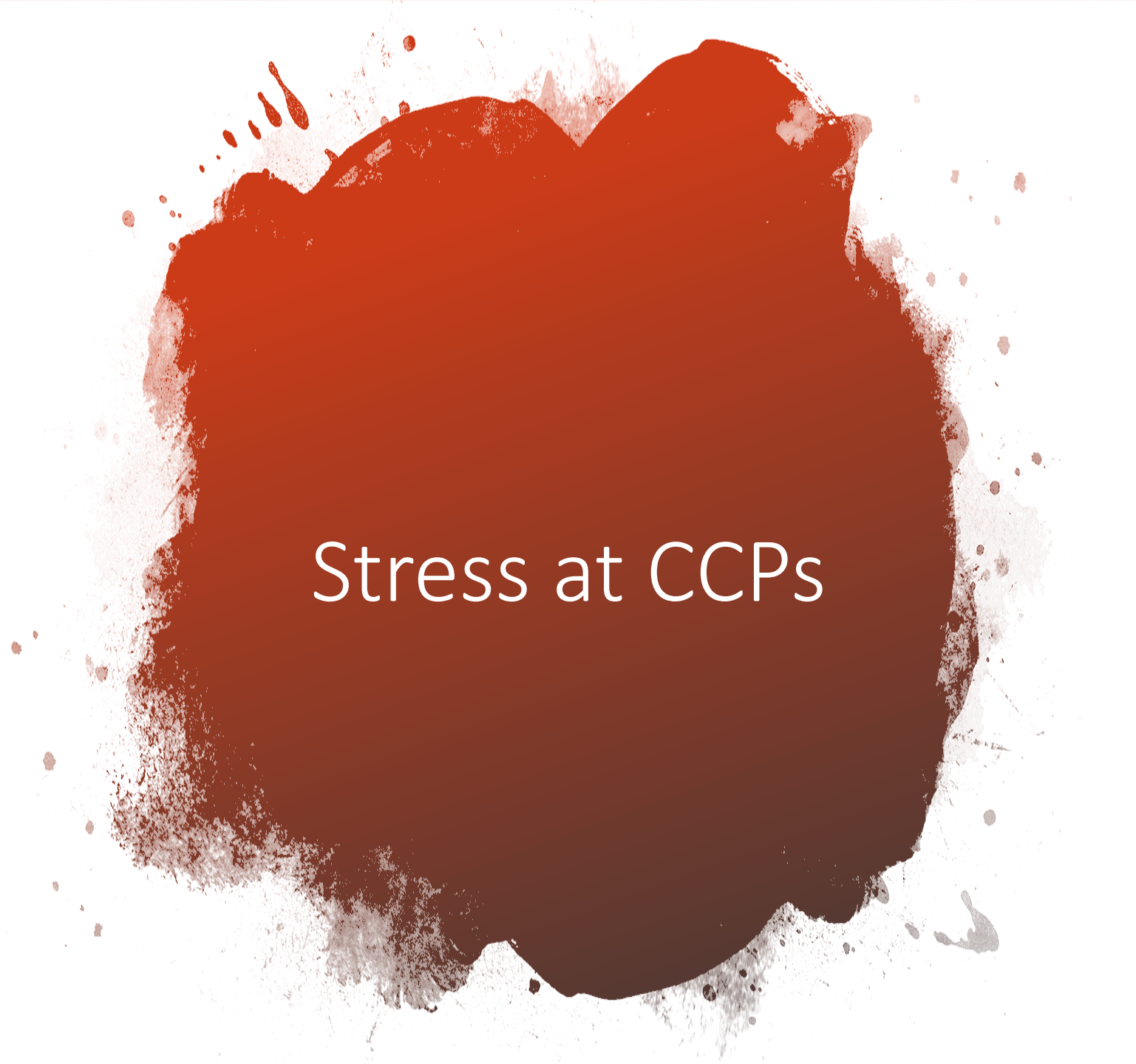
Sinks

- Cash deposited at a central bank of issue
- Cash deposited at authorised credit institutions
- Committed lines of credit or equivalent arrangements with non-defaulting clearing members
- Committed repurchase agreements
- Highly marketable financial instruments that the CCP can demonstrate are readily available and convertible into cash on a **same-day basis** using prearranged and highly reliable funding arrangements, including in stressed market conditions

- Settlement banks
- Payments systems
- Securities settlement systems
- Nostro agents
- Custodian banks
- Liquidity providers
- Interoperating CCPs
- Service providers

Liquidity risk and cover 2 liquidity

- For CCPs, liquidity risk really has two components
 - The usual, **positive liquidity risk** of not having ready access to needed cash in the right currency
 - **Negative liquidity risk**, the risk of having uninvested cash, perhaps that you still have to pay a return on
- The regulatory requirement on positive liquidity is a **cover 2** requirement. The CCP must cover
 - “The liquidity risk generated by the default of at least the two clearing members to which it has the largest exposures”
- EMIR has a ‘95% secured’ requirement
 - “Where cash is maintained overnight ... then not less than 95 % of such cash, calculated over an average period of one calendar month, shall be deposited through arrangements that ensure the collateralisation of the cash with highly liquid financial instrument”
- This means that CCPs either have to use central bank deposit accounts or they have dependence of repo
 - E.g. Eurex (Q3 2018), € 21.5B deposited at central banks in the currency of issue. (Eurex is a credit institution.)



Stress at CCPs

English
Proverb

Don't keep all your eggs in one basket

But...

If you are going to keep all your eggs in one basket, then you had better keep your eye on that basket

How can CCPs become stressed?

Default risk

- More than two defaults
- One or two defaults but with losses beyond cover 2
- Failure to re-balance the book
- Failure of clearing members to perform on their contractual obligations
- Incomplete recovery plan

Liquidity risk

- Running out of cash

Operational and other risks

- Investment risk
- Loss of margin or positions, including failure of a custodian
- Failure of a settlement bank or other connected FMI
- Legal risk
- Cyber risk
- Business risk eroding capital

Recovery

- **Recovery** is the 1st phase of stabilising a stressed financial institution
- For a CCP, it and its members must formulate plans which are agreed in advance and which can be implemented if and when necessary
- Distinguish three important things which the CCP and its members need to achieve
 - Rebalancing the book
 - Absorbing losses
 - Re-resourcing the CCP

Resolution

- If the recovery phase does not, or seems unlikely to, stabilise the CCP then a **resolution authority** ('RA') will step in with its resolution tools
- Triggers for resolution
 - The CCP is failing or likely to fail (perhaps because it has lost the market's confidence)
 - It is reasonably unlikely actions other than resolution (for instance the CCP's own recovery actions or actions directed by its supervisory authority) will restore stability
 - Resolution is in the public interest and in the interests of financial stability

CCP Recovery

- **Default fund calls** require members to pay in additional default fund after their original contributions have been used to absorb losses
- **Variation margin gains haircutting** reduces positive variation margin pay-outs to members
- **Tear up** rebalances the book by eliminating positions
 - In partial tear up, opposite the defaulter
 - In full tear up, the whole service

Tear up (or 'invoicing back')

- The CCP terminates the opposite side of the defaulter's positions, at a fixed potentially off-market price and thus can always rebalance the book
 - E.g., defaulter is short 80 contracts; you are long 10, the rest of the longs are 190; the market price is £10
 - Suppose the defaulter is 40% of the market
 - The CCP determines the IB price is £9, and invoices back pro rata
 - You now own 6 contracts and the CCP pays you £36 for the 4 that you have lost, so you lose £4
- The key issue is the **tear up price**
- Also often available in cases of *force majeure* or 'disorderly markets'



Historical example

Einar Aas and Nasdaq

- Einar Aas was an individual trader who had been very successful in the energy markets
- His net worth was estimated at €2B: he was Norway's largest individual taxpayer in 2016
- He was a self-clearing member of Nasdaq's commodity clearing service and in September 2018 he had a large position on the spread between German and Nordic power

Trouble for Aas

- Nordic power is mostly hydro-electric, and it rained a lot in September, making it cheaper
- German power is mostly fossil-fuel generated, and carbon credits became more expensive in the same period, driving its price up
- Aas's position, based on the spread between the two suffered
- On 10th September, Nasdaq issued an intraday variation margin call, which he did not meet
- He was placed into default on 11th September

Date	Spread change
3 rd September	€ 0.93
4 th September	€ 0.4
5 th September	€ 0.52
6 th September	€ 0.33
7 th September	€ 0.47
10 th September	€ 5.56

17 standard deviation move, "to a level not observed since the Swedish power market was divided into four zones in 2011"

Marginal Protection

- Nasdaq's margin model was calibrated to 99.2% of two day moves **over a one year period**
 - The minimum MPOR for ETDs is 2 days
- In practice this corresponded to a spread change of around €4
- There was no additional concentration margin charged for Aas' position, and the position limit was **50% of the open interest**

Financial resources in the CCP

- The default fund sizing scenarios did not include a large move in the Nordic/German power spread
- Aas was not one of the cover 2 members
- The CCP had a resources structure with both individual service SITG and DF and cross service 'senior capital' and mutualized resources

Resource	Amount
Commodities SITG	€ 7M
Commodities DF	€ 166M
Senior capital	€ 19M
Mutualised DF	€ 45M

- Three other clearing services in addition to commodities:
 - Equity
 - Fixed income
 - OTC rates

Default management and its consequences

- Two auctions were held on Tuesday 10th and Wednesday 12th, of which the latter was successful
- The loss in excess of IM was €114M, so there was a hit to DF of €107M, or 68%
- Additional DF was called from members
- Nasdaq voluntarily injected additional capital to bring the commodities SITG to €19M

Issues

- The CCP financial resources model relies on position risk being covered in IM or DF
- Aas' risk was not fully covered in either place
- Lack of concentration charges

“My position was too large in relation to the liquidity of the market”
- No incentives to bid in the auction (e.g. juniorisation)



Questions